

Discrete mathematics

Dr. D.P. Vidyarthi Room-1

29/11/2022

① Set Theory - Cardinality, Operations, rules, DeMorgan's law, Ordered pairs

② Graph Theory - Vertices and Edges

③ Abstract Algebra -

④ Relations:-

ⓐ $M = \{(x, y) : x \text{ is married to } y\}$

ⓑ $R = \{(x, y) : \text{language } x \text{ is available on computer } y\}$

⑤ Functions:

⑥ Venn diagrams

⑦ Lemma, Theorem, Conjecture.

⑧ Different methods of proving.

⑩ Propositional logic.

Computer Fundamentals:-

Dr. Poonam Agarwal

Room-1

① Digital gadget $\rightarrow$ why binary is used?

② Why CMOS is used over anything else in gates?

③ programming:- Room no-2 Dr. P.P Singh

30/11/2022

50 - End term

50 - Assignments + Mini projects

- Hardware (CPU)
- ↓
- Linkers, loaders, compilers, OS (System software)
- ↓
- Application software
- ↓
- User

Discrete Mathematics:- Trembley Manohar (Book)

01/12/2022

① Relation - Subset of Cartesian product of two sets.

$$R \subseteq A \times B$$

$$(a, b) \in R$$

$$a R b$$

② Properties of relation:- R on set A

- ① Reflexive if $a \in A \forall a (a R a)$
- ② Symmetric $\forall a, b$ (if $a R b$ then $b R a$)
- ③ Transitive $\forall a, b, c$ (if $a R b$ and $b R c$ then $a R c$)
- ④ Asymmetric $\forall a, b$ (if $a R b$ then $b \not R a$)
- ⑤ Antisymmetric $\forall a, b$ (if $a R b$ and $b R a$ then $a = b$)

⑥ Compatibility relation:-

- Ⓐ Reflexive
- Ⓑ Symmetric

⑦ Partial Order relation $[\leq]$

- Ⓐ Reflexive
- Ⓑ Antisymmetric
- Ⓒ Transitive

[Maths notation]

e.g. $\leq$ relation.

⑧ Equivalence relation:- $[=]$

- Ⓐ Reflexive
 - Ⓑ Symmetric
 - Ⓒ Transitive
- e.g. Parallel lines, equal to $(=)$ relation

* Equivalence relation is a compatible relation but the converse may not be true.

③ Examples:-

① $X = \{\text{ball, bed, dog, let, egg}\}$

$R = \{(x, y) : x, y \in X \text{ and } x R y \text{ if } x \text{ and } y \text{ contains some common letters}\} = \{(\text{ball, bed}), (\text{ball, let}), (\text{bed, dog}), (\text{bed, egg}), \dots\}$

→ Reflexive
→ Symmetric } - Compatible not equivalence.

② $a R b \text{ iff } a-b \text{ is divisible by } 3.$

→ Reflexive

→ Symmetric.

→ Transitive

Hence, equivalence.

$$a-b=3k$$

$$b-c=3d$$

$$a-a=3k$$

$$\Rightarrow -(b-a)=3k$$

$$\Rightarrow (b-a)=3(-k)=3n$$

$$(a-b)+(b-c)=3(k+d)$$

$$\Rightarrow a-c=3(k+d)$$

③ $a R b \text{ iff } a > b$

Not equivalence or partial order.

④ $a R b \text{ iff } a+b=8$

⑤ $a R b \text{ iff } a+b < 8$

⑥ $X = \{1, 2, 3, 4\}$

$R = \{(x, y) : x > y\}$

$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$

* $G = (V, E)$

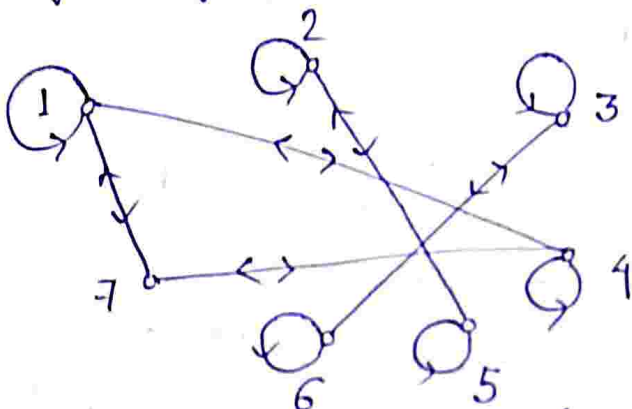
Relation is represented by directed graphs.

e.g. $\begin{array}{ccc} 2 & & 1 & & 3 \\ \circ & \xrightarrow{\quad} & \circ & \xleftarrow{\quad} & \circ \end{array}$

(vii) $X = \{1, 2, 3, \dots, 7\}$

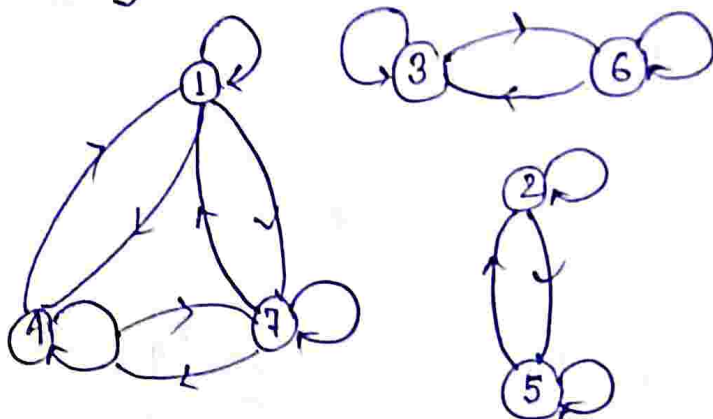
Equivalence relation

$R = \{(x, y) : x - y \text{ is divisible by } 3\}$



$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7),$
 $(4, 1), (5, 2), (6, 3), (7, 4), (7, 1)\}, (1, 7), (2, 5), (3, 6), (4, 7),$
 $(1, 4)\}$

3 classes:



(i) $(1, 1), (4, 4), (7, 7),$
 $(1, 7), (7, 1), (1, 4), (4, 1),$
 $(7, 4), (4, 7)$

(ii) $(3, 3), (6, 6), (6, 3), (3, 6)$

(iii) $(2, 2), (5, 5), (2, 5), (5, 2)$

Equivalence classes

* If an equivalence relation divides the set into disjoint classes then it is called equivalence classes.
Equivalence class partitions the set.

* Let, S be a set and

$A = \{A_1, A_2, \dots, A_n\}$ where each A_i for $i=1, 2, \dots, n$ is a subset of S and the union of A_i 's for $i=1, 2, \dots, n$ is equal to S , i.e., $\bigcup_{i=1}^n A_i = S$.

Now, if the elements of A_i are disjoint then A is called partition of S .

$A_1, A_2, \dots, A_n$ are called the blocks of the partition.

Otherwise, all A_i 's are called the covering of S .

[i.e, if A_i 's are not disjoint]

④ Examples:-

① $S = \{a, b, c\}$

$A = \{\{a, b\}, \{b, c\}\}$

$C = \{\{a\}, \{b, c\}\}$ $E = \{\{a\}, \{b\}, \{c\}\}$

$B = \{\{a\}, \{a, c\}\}$

$D = \{\{a, b, c\}\}$

$F = \{\{a\}, \{a, b\}, \{a, c\}\}$

A, B, C, D, E, F are sub-sets of sub-sets of S .

Covering - A, F

Partition - C, D, E

① $A = \{1, 2, 3, 4\}$

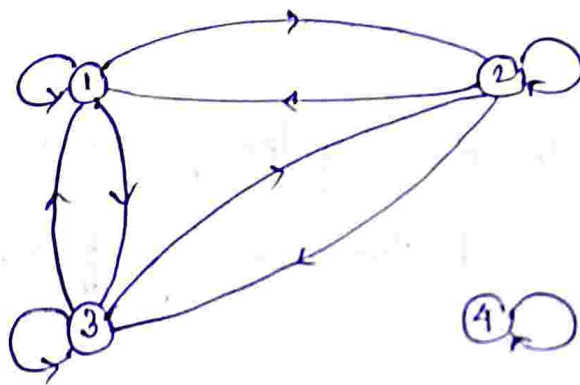
$R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,2), (3,1), (4,1), (3,3)\}$

① Reflexive - $(1,1), (2,2), (3,3), (4,4)$

② Symmetric -

③ Transitive -

$EQ = \{\{1,2,3\}, \{4\}\}$



It is an equivalence relation and forms equivalence classes.

* What calls main ?

* Macros ? Preprocessor ?

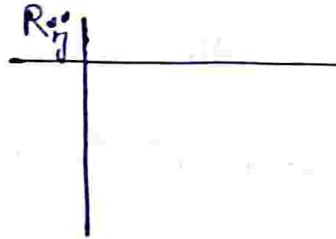
* pointer ?

13/12/2022

Discrete Mathematics:-

- ① $R \subseteq A \times B$ where A, B are two non-empty sets.
- ② Relation can be represented using the matrix.
- ③ Relation matrix:-

$$r_{ij} = \begin{cases} 1 & \text{if } x_i R y_j \\ 0 & \text{otherwise} \end{cases}$$



④ Composition of relations:-

Let, R and S be two relations, then

$$R \circ S = \{(x, z) : x \in X \wedge z \in Z \wedge \exists y \in Y \wedge (x, y) \in R \wedge (y, z) \in S\}$$

i.e, If $x \rightarrow y$ and $y \rightarrow z$ then $x \rightarrow z$.

e.g. $R = \{(1, 2), (3, 4), (2, 2)\}$ and $S = \{(4, 2), (2, 5), (3, 1)\}$

$$R \circ S = \{(1, 5), (3, 2), (2, 5)\}$$

$$S \circ R = \{(1, 2), (3, 2)\}$$

$\mathbb{Z} = \mathbb{I}$

e.g. $R = \{(x, 2x) : x \in \mathbb{Z}\}$ and $S = \{(x, 7x) : x \in \mathbb{I}\}$

$$R \circ S = \{(x, 14x) : x \in \mathbb{Z}\} = S \circ R \quad R \circ R = \{(1, 2), (2, 4), (3, 6), (4, 8), \dots, (n, 2n)\}$$

$$R \circ R = \{(x, 4x) : x \in \mathbb{Z}\} = \{(1, 4), (2, 8), \dots, (n, 4n)\}$$

$$R \circ S = \{(1, 14), (2, 28), (3, 42), \dots, (n, 14n)\}$$

$$S \circ R = \{(1, 14), (2, 28), (3, 42), \dots, (n, 14n)\}$$

$$S = \{(1, 7), (2, 14), (3, 21), (4, 28), \dots, (n, 7n)\}$$

⑤ Propositional Logic:

① Premises builds to conclusion in a proof:

② Consider statements that are either true or false not in between (mathematically interesting statements are to be considered).

③ Law of excluded middle.

④ Proposition is a statement (mathematical) which is either true or false.

⑤ A theorem is a proposition which has been proved to be true about a mathematical system.

⑥ It is not the case that 5 is a prime no.

⑦ $\{p\}$ means this particular statement is a proposition.

⑧ p : if x is an integer then x^2 is positive integer.

⑨ Capital alphabet usually use as predicate

⑩ Connective : AND, OR, NOT used to generate new proposition.

• AND: $p \wedge q \Rightarrow p \text{ and } q$, here $\wedge$ represents conjunction.

• OR: $p \vee q \Rightarrow p \text{ or } q$, here $\vee$ represents disjunction.

• NOT: $\neg p \Rightarrow \text{not } p$, here $\neg$ represents negation.

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

p	$\sim p$
T	F
F	T

e.g. Determine whether $((x=3) \text{ and } (y < 9)) \text{ OR } (y \geq 4)$
AND $(\text{NOT } ((x=3) \text{ AND } (y \geq 4)))$ is true or false if $x=2, y=10$.

$\sim p$

p q r
T F T

$$\rightarrow ((p \wedge q) \vee r) \wedge (\sim (p \wedge r))$$

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$	$p \wedge r$	$\sim (p \wedge r)$	$((p \wedge q) \vee r) \wedge (\sim (p \wedge r))$
T	T	T	T	T	T	F	F
T	T	F	T	T	F	T	T
T	F	F	F	F	F	T	F
F	T	T	F	T	F	T	T
F	T	F	F	F	F	T	F
F	F	T	F	T	F	T	T
F	F	F	F	F	F	T	F
T	F	T	F	T	T	F	F

Assignment:-

e.g. p : A byte has 7 bits F Determine $p \wedge q$, $p \vee q$, $\sim p$
 q : A word is 2 bytes T $\sim(p \wedge q)$, $(\sim p \vee \sim q)$,
 r : A bit is a 0 or 1 T

p	q	r	$\sim p$	$p \wedge q$	$p \vee q$	$\sim(p \wedge q)$	$\sim q$	$(\sim p \vee \sim q)$
T	T	T	F	T	T	F	F	F
T	T	F	F	T	T	F	F	F
T	F	T	F	F	T	T	T	T
T	F	F	F	F	T	T	T	T
F	T	T	T	F	T	T	F	T
F	T	F	T	F	T	T	F	T
F	F	T	T	F	F	T	T	T
F	F	F	T	F	F	T	T	T

15/12/2022

Discrete Mathematics:-

p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$p \wedge q$	$(p \wedge q) \vee (\sim p \vee \sim q)$
F	T	T	F	T	F	T
F	F	T	T	T	F	T
T	T	F	F	F	T	T
T	F	F	T	T	F	T

* Tautology
(means all final conclusion are true for the statement)

• Commutative property:-

$$\rightarrow p \wedge q = q \wedge p$$

$$\rightarrow p \vee q = q \vee p$$

• Associative property:-

$$\rightarrow (p \vee q) \vee r = p \vee (q \vee r)$$

$$\rightarrow (p \wedge q) \wedge r = p \wedge (q \wedge r)$$

• Distributive property:-

$$\rightarrow p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

$$\rightarrow p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$$

• Law of Duality:-

2 statements ^{is} said to be dual if ^{conjunction} conjunction is converted into disjunction and vice-versa. $T \rightarrow F$ and $F \rightarrow T$.

$$\bullet (p \vee q) \wedge r \xleftrightarrow{\text{dual}} (p \wedge q) \vee r$$

$$\bullet (p \wedge q) \vee F \xleftrightarrow{\text{dual}} (p \vee q) \wedge F$$

$$\ast (p \wedge q \vee r) \wedge \cup (p \wedge r) = (\cup p \wedge r) \vee [(p \wedge q) \wedge \cup r] \text{ [Proof if]}$$

• Disjunction of conjunction:

(i) if $(x=0)$ OR $[\text{NOT } ((x>7) \text{ OR } (a=5))]$

(ii) NOT $[(x=0) \text{ OR } (b=7)] \text{ OR } (a<5)$

(iii) $(x=7)$ OR $[\text{NOT } ((a \leq 5) \text{ AND } (x=0))]$

• DeMorgan's Law:-

$$\rightarrow \sim(p \wedge q) = \sim p \vee \sim q$$

$$\rightarrow \sim(p \vee q) = \sim p \wedge \sim q$$

$$\bullet p \vee p = p$$

$$\bullet \cup \cup p = p$$

$$\bullet (p \wedge \cup p) \vee q = q$$

$$\bullet p \vee \cup p = q \vee \cup q$$

- $p \rightarrow q$ (" $\rightarrow$ " conditional operator) (only if)

$p \rightarrow q \neq q \rightarrow p$ i.e., converse ~~not true~~ of each other but not same
 $p \rightarrow$ antecedent $q \rightarrow$ consequent $\rightarrow$ implication operator

p	q	$p \rightarrow q$	$q \rightarrow p$	consequent implication operator			$\neg(p \wedge \neg q)$	$\neg q \rightarrow \neg p$
T	T	T	T	T	T	T	T	T
T	F	F	T	F	F	F	F	F
F	T	T	F	T	T	T	T	T
F	F	T	T	T	T	F	T	T

$$\neg q \rightarrow \neg p$$

$\neg q \rightarrow \neg p$ contrapositive of $p \rightarrow q$

$$p \rightarrow q = \neg p \vee q$$

$$p \rightarrow q = \neg q \rightarrow \neg p$$

contrapositive

$p \rightarrow q$ inverse $\neg p \rightarrow \neg q$ $p \rightarrow q = \neg(p \wedge \neg q)$

converse $q \rightarrow p$

- Converse and inverse are equivalent to each other but not equivalent to the actual statement.
- If p then q does not mean p causes q to be true.
- $(p \rightarrow q)$ is converse of $(q \rightarrow p)$ and vice-versa.

• Biconditional Statements $\leftrightarrow$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

→ (if p then q and if q then p) (if and only if)

if to prove two statements are equivalent then you can have bi conditional of two statements you get tautology.

Then they are equivalent to each other.

For e.g. $\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$

* Logical Collection of statements logically implies another collection of statements if whenever all the statements in the first collection are true then all the statements in the second collection must be true and known as rule of inference.

$$\textcircled{i} \quad p \text{ if } (x=0) \text{ OR } [\text{NOT } (x>7) \text{ OR } (a=5)]$$

$\downarrow$ $\downarrow$ $\downarrow$
 p q r

$$\rightarrow p \vee [\sim(q \vee r)]$$

$$\rightarrow p \vee [\sim q \wedge \sim r]$$

$$\rightarrow (p \vee \sim q) \wedge (p \vee \sim r)$$

$$\textcircled{ii} \quad \text{NOT } [(x=0) \text{ OR } (b=7)] \text{ OR } (a<5)$$

$\downarrow$ $\downarrow$ $\downarrow$
 p q r

$$\rightarrow \sim [(p \vee q) \vee r]$$

$$\rightarrow \sim(p \vee q) \wedge (\sim r)$$

$$\rightarrow (\sim p) \wedge (\sim q) \wedge (\sim r)$$

$$\textcircled{iii} \quad (x=7) \text{ OR } [\text{NOT } (a \leq 5) \text{ AND } (x=0)]$$

$\downarrow$ $\downarrow$ $\downarrow$
 p q r

$$\rightarrow p \vee [\sim(q \wedge r)]$$

$$\rightarrow p \vee (\sim q \vee \sim r)$$

$$\rightarrow p \vee (\sim q) \vee (\sim r)$$

Discrete Mathematics:-

Mid Sem - After winter break 29/12/2022

Rules of Inference:-

$A_1, A_2, \dots, A_n = \text{Arguments.}$

$A_1, A_2, \dots, A_{n-1}$ proves A_n

premises

$(A_1 \wedge A_2 \wedge A_3 \dots \wedge A_{n-1}) \rightarrow A_n$

$$P \wedge T \equiv P$$

$$P \vee T \equiv T$$

$$P \wedge F \equiv F$$

$$P \vee F \equiv P$$

Conclusion, $P \wedge \neg P \equiv F$

It gives $P \vee \neg P \equiv T$

Tautology, then this is a valid argument.

• $p \rightarrow q$, p be true then q must be true i.e.,

p	q	$p \rightarrow q$	$(p \rightarrow q \wedge p) \rightarrow q$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

$[p \rightarrow q, p \vdash q]$ it means, premises $p \rightarrow q$ and p proves q . Rule - MODUS PONENS (method of affirming)

e.g. p : Sheila solves seven problems correctly.

q : Sheila obtained grade A.

$p \rightarrow q$: If Sheila solves seven problems correctly, then she gets grade A.

If $p \rightarrow q$, p true, then q must be true.

e.g. p : Ram solves 17 problems correctly.

q : Ram obtains grade A.

If $p \rightarrow q$, q true, then p ~~must be true~~ might not be true.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge q$	$[(p \rightarrow q) \wedge q] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

i.e., $p \rightarrow q, q \not\vdash p$

$p \rightarrow q, \neg q \vdash \neg p$ [Rule of inference - MODUS TOLLENS]

p	q	$\neg p$	$\neg q$	$(p \rightarrow q) \wedge (\neg q)$	$[(p \rightarrow q) \wedge (\neg q)] \rightarrow (\neg p)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	T

Fallacies: When premises are true but conclusion is false

• $p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$ Rule of inference: Hypothetical Syllogism

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	$p \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	F
T	F	T	F	T	F	T	T
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

• $p \vee q, \neg p \vdash q, p \vee q, \neg q \vdash p$ Rule of inference: Disjunctive Syllogism

p	q	$\neg p$	$\neg q$	$p \vee q$	$((p \vee q) \wedge \neg p) \rightarrow q$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	F	F

• $p \wedge q \vdash p, p \wedge q \vdash q$ Rule of inference: SIMPLIFICATION

• $p \vdash p \vee q$ Rule of inference: ADDITIONS

• $p, q \vdash p \wedge q$. Rule of inference: Conj CONJUNCTION •

• $p \rightarrow q, \neg q \vdash \neg p$ Rule of inference: CONTRADICTION
 $\neg q \rightarrow \neg p, \neg q \vdash \neg p$ [Modus ponens] DS Mallick
MK Sen .
 $\rightarrow p \rightarrow q, \neg q \vdash \neg p$ [by ~~con~~ taking contrapositive
[proved] of $\neg q \rightarrow \neg p$]

Q. Prove that $[(p \rightarrow q) \wedge (\neg r \rightarrow \neg q) \wedge \neg r] \rightarrow \neg p$ is a Tautology.

1. $p \rightarrow q$ (given)

2. $\neg q \rightarrow \neg p$ (Contrapositive)

3. $\neg r \rightarrow \neg q$ (given)

4. $\neg r \rightarrow \neg p$ [hypothetical Syllogism]

5. $\neg r$ (given)

6. $\neg p$ (Modus ponens)

Q. $p \vee q, \neg p \vee r$ prove $p \vee r$.

1. $p \vee q$ (given)

2. $\neg p \rightarrow q$ (equivalent)

3. $\neg q \vee r$ (given)

4. $q \rightarrow r$ [equivalent]

5. $\neg p \rightarrow r$ [hypothetical Syllogism]

6. $p \vee r$ [equivalent]

H.W

① $p, p \rightarrow q, q \rightarrow r$ prove r

② $p \vee q, \neg p$ prove q

③ $p \wedge \neg q, p \rightarrow r, r \rightarrow (s \vee q)$ prove s .

④ $p, q, (p \wedge q) \rightarrow r$ prove r .

⑤ $p \leftrightarrow q, \neg p \rightarrow r, \neg r$ prove q

⑥ $p \vee q, \neg q \vee r$ prove $p \vee r$

⑦ $p \leftrightarrow q, q \leftrightarrow r$ prove $p \leftrightarrow r$

⑧ $p, (p \wedge \neg q) \rightarrow \neg p$ prove $p \leftrightarrow q$

• Predicate Logic:-

A statement which would be a proposition except for the fact that it includes variables whose values are not specified is called predicate.

e.g. (i) If $x < 3$ then $x^2 = -1$: $P(x)$ One-place predicate.

(ii) Sum of first n odd integers is n^2 : $Q(n)$

(iii) $x+4=3$ the value of x determines if it is a proposition or a predicate.

(iv) If $x < y$ then $x^2 < y^2$: $S(x, y)$ Two-place predicate.

(v) $x+y=4$ if and only if $y=4-x$

If we quantify the variable then predicate becomes proposition.

Predicate are generally denoted by Capital Letters. $P, Q, \dots$

Proposition are generally denoted by small letters. $p, q, \dots$

① $p, p \rightarrow q, q \rightarrow r$ proves r

① p [Given]

② $p \rightarrow q$ [Given]

③ $q \rightarrow r$ [Given]

④ $p \rightarrow r$ [hypothetical syllogism]

⑤ r [Modus ponens]

② $p \vee q, \sim p$ prove q .

① $p \vee q$ (Given)

② $\sim p$ (Given)

③ q (Disjunctive syllogism)

③ $p \wedge \sim q, p \rightarrow r, r \rightarrow (s \vee q)$ prove s

Discrete Mathematics:-

23/12/2022

- * An n place predicate is said to be satisfiable provided that there is an n -tuple which satisfies it. And if all n -tuples satisfy it, then it is said to be the valid predicate.
- * Two predicates are said to be equivalent if we take the biconditional of them and get tautology.

e.g. (a) $a+b = b+a, \forall a, b$

(b) $a+(b+c) = (a+b)+c, \forall a, b, c$

(c) $a+0 = 0+a = a, \forall a$

(d) For every $a, \exists$ a b such that $a+b=0$ and $b+a=0$.

(e) If $a < b$ then there is a c such that c is positive and $a+c=b$.

(f) If $a < b$ and $b < c$ then $a < c$.

Let, $P(a, b): a+b = b+a, \forall a, b$ [Universal quantifier $\rightarrow \forall$]

$Q(a, b, c): a+(b+c) = (a+b)+c, \forall a, b, c$

In (d), the value of b is existential quantifier $\exists$.

$S(a, b): \forall a \exists b \text{ s.t. } a+b=0 \text{ and } b+a=0.$

- $\forall a \exists b S(a,b)$ means for all values of a , there is some b , for which $S(a,b)$ is true.
- But, $\exists b \forall a S(a,b)$ means for some b $S(a,b)$ is true irrespective of the value of a .

③ If $a < b$ then $\exists c$ s.t. $c > 0$ and $a+c=b$.

$$P(a,b) : a < b, \exists c (c > 0), a+c=b$$

$$\forall a \exists b \{ P(a,b) \rightarrow \exists c [Q(c) \wedge R(a,b,c)] \}, \text{ here } Q(c) : c > 0$$

$$R(a,b,c) : a+c=b$$

$$P(a,b) : a < b$$

$$④ \quad \forall a (P(a,b) \wedge Q(b,c)) \rightarrow R(a,c), \forall a \forall b \forall c$$

Bound variable $\rightarrow \forall a$ Proposition is predicate if
 free variable $\rightarrow \forall b$ variable is bound.

Q. $P(x,y) : x$ is older than y .

$Q(x,y) : x$ is wiser than y .

$R(x) : x$ is bald.

$$\forall x \exists y [P(x,y) \rightarrow [Q(x,y) \wedge R(y)]]$$

$\rightarrow$ For every x , there is y such that x is older than y then x is wiser than y and y is bald.

Q. ② $\forall x \forall y [Q(x, y) \rightarrow R(y)]$

Q. ③ $\forall x [\forall y \in P(x, y) \rightarrow R(x)]$

② For every x and y if x is wiser than y then y is bald.

③ For every x if x is not older than y for all y , then x is bald. This is true for all x .

Q. ④ $\sim \forall x P(x)$ is equivalent to $\exists x \neg P(x)$

eg. $P(x)$: x is taller than 1 mt.

Q. ⑤ $\sim \exists x P(x)$ is equivalent to $\forall x \neg P(x)$

no one is taller than 1 mt.

Q. ⑥ $P(x, y)$: x is faster than y .

$Q(x, y)$: y is taller than x .

$R(x)$: x is more than 70 kg

$P(x, Ram)$: x is faster than Ram.

All are faster than Ram.

$Q(Ram, Mohan) \wedge R(Ram)$: Ram is taller than Mohan and Ram is more than 70 kg.

$P(x, y) \rightarrow Q(x, y)$ If x is faster than y then y is taller than x .

$Q(x, y) \rightarrow \neg R(x)$, If y is taller than x then x is not more than 70 kg.

$P(\text{Ram}, \text{Mohan}) \vee [Q(\text{Ram}, \text{Mohan}) \wedge R(\text{Mohan})]$:

Ram is faster than Mohan or

- $\forall x P(x)$, Let there be only two universal values of x i.e. a and b , then, $P(a)$ is true and $P(b)$ is also true.

$$\therefore \forall x P(x) = P(a) \wedge P(b)$$

- $\exists x P(x)$, Let, there be some two values of x i.e. a and b , then, $P(a)$ may be true or $P(b)$ may be true.

$$\therefore \exists x P(x) = P(a) \vee P(b)$$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\forall x P(x) = P(a) \wedge P(b)$$

$$\exists x P(x) = P(a) \vee P(b)$$

Discrete Mathematics:-

27/12/2022

- A statement formula = $A(p_1, p_2, \dots, p_n)$ where p_i is either a proposition or predicate.
The no. of length of truth table = 2^n .

• Elementary product:-

- Elementary sum:- Same as elementary product but with disjunction operator.

Let, p and q be two variables, so elementary product will be defined as a ~~variable~~ product of the variable and their negation i.e., conjunction of variables and their negation. These are said to be elementary product.

- e.g. $p \wedge \sim p$ $\sim p \wedge q$, here $p, \sim p$ are factors.

- e.g. $p \vee \sim q$.

- A necessary and sufficient condition for an elementary product to be identically false is that it contains at least one pair of factors in which one is the negation of the other. i.e. iff $p \wedge \sim p$ is false.

- A necessary and sufficient condition for an elementary sum to be identically true is that ~~if~~ it contains at least one pair of factors in which one is the negation of the other. i.e. iff $p \vee \sim p$ is True.

• $A(p, q) = (p \wedge \sim q) \vee (q \wedge p)$ disjunctive normal form.

• $B(p, q) = (p \vee \sim q) \wedge (q \vee p)$ conjunctive normal form.

① Example:- $p \wedge (q \vee \sim r)$ Convert to disjunctive normal form

$$= (p \wedge q) \vee (p \wedge \sim r)$$

$$② [(p \wedge q) \wedge (r \vee \sim r)] \vee [(p \wedge \sim r) \wedge (q \vee \sim q)]$$

$$③ (p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \vee (p \wedge \sim r \wedge q) \vee (p \wedge \sim r \wedge \sim q)$$

④ • Procedure:-

① If you have the connectives like conditional and biconditional in the statement formula then obtain an equivalent form containing conjunction, disjunction and negation.

$$p \rightarrow q = \sim p \vee q = p \vee \sim q$$

$$② p \wedge (p \rightarrow q) \\ = p \wedge (\sim p \vee q) \\ = (p \wedge \sim p) \vee (p \wedge q),$$

required disjunctive normal form.

$$③ \sim(p \vee q) \leftrightarrow (p \wedge q)$$

$$= [\sim(p \vee q) \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow \sim(p \vee q)]$$

$$= [\sim(p \wedge \sim q) \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow (\sim p \wedge \sim q)]$$

$$= [\sim(\sim p \wedge \sim q) \vee (p \wedge q)] \wedge [(p \wedge q) \vee \sim(\sim p \wedge \sim q)]$$

$$= [(p \vee q) \vee (p \wedge q)] \wedge [(p \wedge q) \vee (p \vee q)]$$

- $A(p, q) = (p \wedge \sim q) \vee (q \wedge p)$ disjunctive normal form.
- $B(p, q) = (p \vee \sim q) \wedge (q \vee p)$ ~~conjunctive~~ conjunctive normal form.

① Example:- $p \wedge (q \vee \sim r)$ Convert to disjunctive normal form

$$= (p \wedge q) \vee (p \wedge \sim r)$$

$$② [(p \wedge q) \wedge (r \vee \sim r)] \vee [(p \wedge \sim r) \wedge (q \vee \sim q)]$$

$$③ (p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \vee (p \wedge \sim r \wedge q) \vee (p \wedge \sim r \wedge \sim q)$$

④ • Procedure:-

- ① If you have the connectives like conditional and biconditional in the statement formula then obtain an equivalent form containing conjunction, disjunction and negation.

$$p \rightarrow q = \sim p \vee q \quad p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$$

$$② p \wedge (p \rightarrow q)$$

$$= p \wedge (\sim p \vee q)$$

$$= (p \wedge \sim p) \vee (p \wedge q)$$

required disjunctive normal form.

$$③ \sim(p \vee q) \leftrightarrow (p \wedge q)$$

$$= [\sim(p \vee q) \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow \sim(p \vee q)]$$

$$= [\sim p \wedge \sim q \rightarrow (p \wedge q)] \wedge [(p \wedge q) \rightarrow (\sim p \vee \sim q)]$$

$$= [(\sim p \wedge \sim q) \vee (p \wedge q)] \wedge [(p \wedge q) \vee \sim(p \wedge q)]$$

$$= [(p \vee q) \vee (p \wedge q)] \wedge [(p \wedge q) \vee (\sim p \vee \sim q)]$$

• Principal conjunctive / Disjunctive normal form:-

This is unique for all.

p, q								
p	q	$p \wedge q$	$\neg p$	$\neg q$	$\neg p \wedge q$	$p \wedge \neg q$	$\neg p \wedge \neg q$	$p \rightarrow q$
T	T	T	F	F	F	F	F	T
T	F	F	F	T	F	T	F	F
F	T	F	T	F	T	F	F	T
F	F	F	T	T	F	F	T	T

$\therefore p \rightarrow q \mapsto (p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$ [Principal disjunctive normal form]

$p \vee q$

T

T

T

F

Express $p \vee q$ in principal disjunctive normal form.

$$\therefore p \vee q \mapsto (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$$

$$\neg(p \wedge q) = (\neg p \vee \neg q)$$

F

T

T

T

$\neg p \vee q$

T

F

T

T

$$\neg(p \wedge q) \rightarrow (p \wedge \neg q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

• Obtain the principal disjunctive normal form of

$$\neg p \vee q$$

$$\rightarrow \neg p \vee q$$

$$= [\neg p \wedge (q \vee \neg q)] \vee [q \wedge p \vee (p \vee \neg p)]$$

$$y + \bar{y} = 1$$

$$= (\underbrace{\neg p \wedge q}_T) \vee (\underbrace{\neg p \wedge \neg q}_T) \vee (\underbrace{q \wedge p}_T) \vee (q \wedge \neg p)$$

$$= (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (q \wedge p)$$

• $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$ Obtain the principal disjunctive normal form.

$$= [(p \wedge q) \wedge (r \vee \neg r)] \vee [(\neg p \wedge r) \wedge (q \vee \neg q)] \vee [(q \wedge r) \wedge (p \vee \neg p)]$$

$$= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge r \wedge q) \vee (\neg p \wedge r \wedge \neg q) \vee (q \wedge r \wedge p) \vee (q \wedge r \wedge \neg p)$$

$$= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge r \wedge q) \vee (\neg p \wedge r \wedge \neg q)$$

Discrete Mathematics

Algebra/Algebraic structure 29/12/2022
Boolean Algebra.

• Boolean Algebra:-

→ Boolean algebra is a relation,

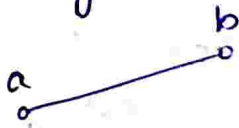
→ Boolean algebra is a partial order relation. $(P, \leq) \Rightarrow$ poset.

- Reflexive
- Anti-symmetric
- Transitive.

• Definition:-

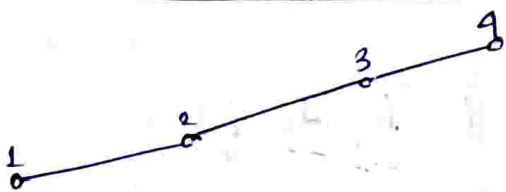
① Let, $(P, \leq)$ be a poset, If $\forall x, y \in P$, we have, either $x \leq y$ $\vee$ $y \leq x$, then $\leq$ relation is said to be simple ordering or linear ordering on P and now, this $(P, \leq)$ is called the totally ordered set or Chain.

Hasse diagram: It is specific to poset.



e.g. $(P, \leq)$ be a poset. where $P = \{1, 2, 3, 4\}$ with $\leq$ relation.

$$(P, \leq) = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

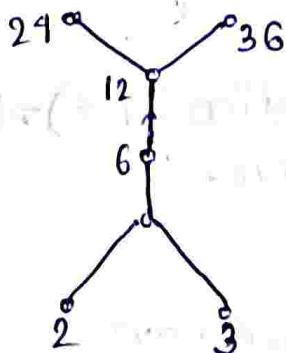


It is a totally ordered set and forms chain.

Hasse diagram

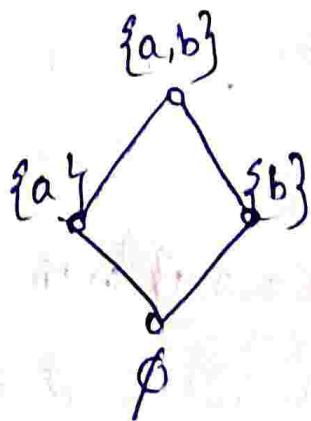
e.g. $X = \{2, 3, 6, 12, 24, 36\}$ and partial order relation $x \leq y$ which says that x divides y .

- ①
- This example is not a chain.
 - $\therefore$ Every partial order relation is not a chain.
 - $(2, 3), (24, 36)$ are incomparable.
 - ~~x covers y~~ y covers x .
 - e.g. 12 covers 6, 6 covers both 2 and 3.



e.g. Relation is $\subseteq$ and the set is $P = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

②



- $\{a\}$ and $\{b\}$ are not comparable.
- $\{a\}, \{b\}$ cover $\emptyset$.
- $\{a, b\}$ covers $\{a\}$ and $\{b\}$.

- ii) Let, R be a relation s.t. $(P, \leq)$ i.e., $R \leftrightarrow (P, \leq)$
 $\tilde{R}$ is converse of R s.t. $(P, \geq)$ i.e., $\tilde{R} \leftrightarrow (P, \geq)$

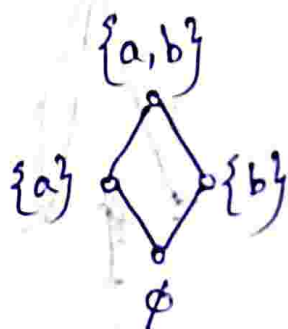
Examples:-

- ① Let, A be a given set and $P(A)$ = power set of A , and the given relation $< P(A), \leq >$ when $A = \{a\}$

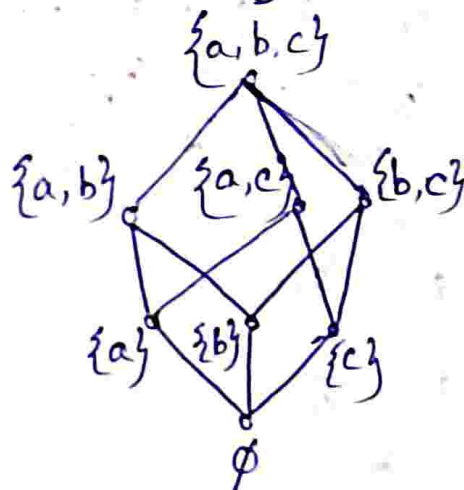


simple chain.

- ② $A = \{a, b\}$ and same as ①:

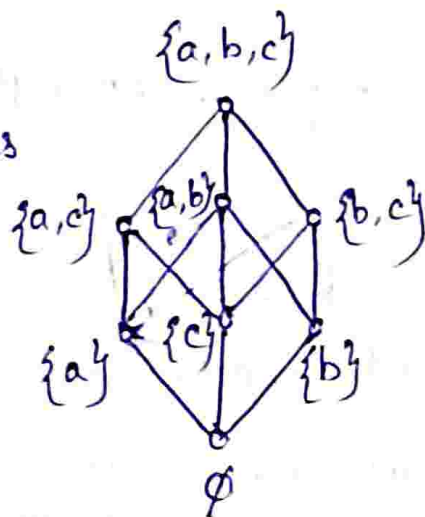


- ③ $A = \{a, b, c\}$ and same as ①.

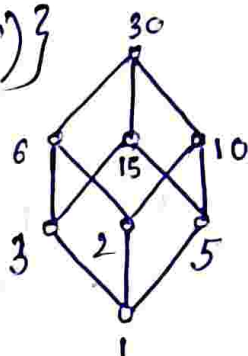
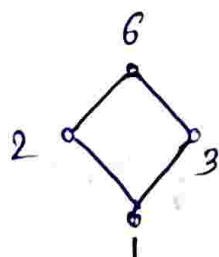


- ④ Let, A = Set of factors of a particular +ve integer m .
 and let, $\leq$ be the relation divides
 s.t. $\leq = \{(x, y) : x \in A \wedge y \in A \wedge (x \text{ divides } y)\}$

When $m = 2$ $\leq = \{1, 2\}$



⑤ $m = 6$ $\leq = \{(1, 2), (1, 3), (2, 6), (3, 6)\}$



⑥ $m = 30$

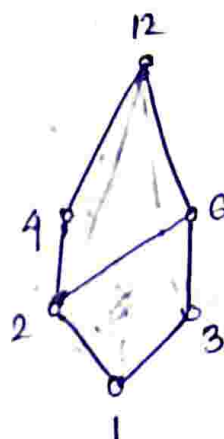
$A = \{1, 2, 3, 5, 6, 10, 15, 30\}$

same as cube.

⑦ $m = 12$

$A = \{1, 2, 3, 4, 6, 12\}$

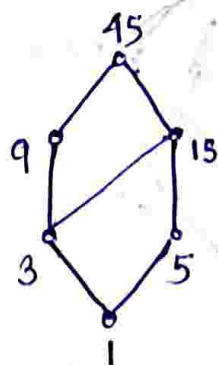
$\leq = \{(1, 2), (1, 3), (1, 4), (1, 6), (1, 12), (2, 4), (2, 6), (2, 12)\}$



⑧ $m = 45$

$A = \{1, 3, 5, 9, 15, 45\}$

$\leq = \{(1, 3), (1, 5), (1, 9), (1, 15), (1, 45), (3, 9), (3, 15), (3, 45), (5, 15), (5, 45), (9, 45)\}$



⑨ If $\exists y \in P$ s.t. $y \leq x \forall x \in P$, then y is called the least member in P .

Similarly, if $\exists y \in P$ s.t. $x \leq y \forall x \in P$, then y is called the greatest member in P .

→ Least and greatest member are unique. The above examples satisfy this condition except 1st one (A)

⑩ Further, we define an element $y \in P$ is called a minimal member of P relative to the partial order relation if for

no $x \in P$ is $x < y$. They need not be unique. All least members are minimal but not vice-versa.
e.g. All ex examples.

• Similarly, maximal member of P related $\leq$,
no $x \in P$ s.t. $y < x$. They need not be unique. All greatest members are maximal but not vice-versa.
e.g. All examples.

te

① $(P, \leq)$ Poset, $A \subseteq P$ Any element $x \in P$ is an upper bound for A if $\forall a \in A, a \leq x$.

Similarly, any element $x \in P$ is a lower bound for A if for all $a \in A, x \leq a$.

e.g. $A = \{a, b, c\} \quad \langle P(A), \leq \rangle$

Let, $B \subseteq P(A) = \{\{b, c\}, \{b\}, \{c\}\}$

Upper bound = $\{b, c\}, \{a, b, c\}$

Lower bound = $\emptyset$

$P(A) = \{\emptyset, \{b\}, \{a\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$

e.g. $C = \{\{a, c\}, \{c\}\}$

Upper bound = $\{a, c\}, \{a, b, c\}$

Lower bound = $\emptyset, \{c\}$

U

17/01/2023 Discrete Mathematics:-

① Let, $\langle P, \leq \rangle$ be a poset, Let, $A \subseteq P$ any element $x \in P$ is an upper bound for A if $\forall a \in A, a \leq x$.

e.g. $A = \{a, b, c\}$ $\langle P(A), \subseteq \rangle$ is a partial order set.

Let, $B = \{\{b, c\}, \{b\}, \{c\}\} \subseteq P(A)$

$\therefore$ Upper bound = $\{b, c\}, \{a, b, c\}$

② Let, $\langle P, \leq \rangle$ be a poset. Let $A \subseteq P$, any element $x \in P$ is a lower bound for A if $\forall a \in A, x \leq a$.

$\therefore$ Lower bound = $\emptyset$

Let $C = \{\{a, c\}, \{c\}\}$

Here lower bound = $\emptyset, \{c\}$

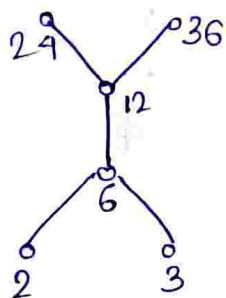
Upper bound = $\{a, c\}, \{a, b, c\}$

e.g. $A = \{2, 3, 6, 12, 24, 36\}$ divides relation.

$$B = \{2, 3, 6\}$$

Upper bound = $\{6, 12, 24, 36\}$

Lower bound = none.



③ Least Upper bound and Greatest lower bound:-

Let $\langle P, \leq \rangle$ be a poset and let $A \subseteq P$, then we define $x \in P$ is a least upper bound for A if x is upper bound and $x \leq y$ where y is any upper bound for A .

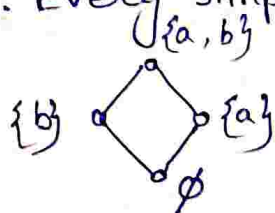
Also called supremum.

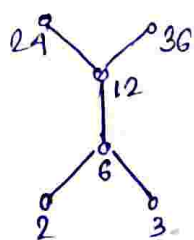
Let $\langle P, \leq \rangle$ be a poset and let $A \subseteq P$, then we define $x \in P$ is a greatest lower bound for A if x is a lower bound and $x \geq y$ where y is any lower bound for A .

Also called infimum.

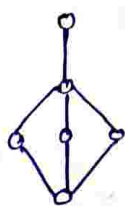
④ Lattice:- Lattice is a poset $\langle L, \leq \rangle$ in which every pair of elements $a, b \in L$ has a least upper bound and a greatest lower bound. (We will consider finite type set only) [as we can't consider the GLB for infinite sets]

e.g. Every simply ordered set/chain is a lattice.

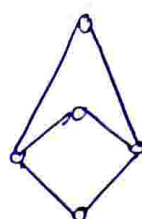




Not a lattice



Lattice

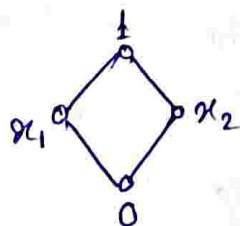


Not a lattice.

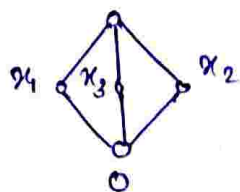
⑤ A Lattice is called a complete lattice if each of its non empty subsets has a LUB and GLB.

⑥ If in a bounded lattice $\langle L, *, +, 0, 1 \rangle$ and element $b \in L$ is called a complement of $a \in L$ if $a * b = 0$ and $a + b = 1$. [Generally in digital electronics $*$ is AND operation and $+$ is OR operation]

⑦



x_1 and x_2 are complement of each other.



x_1, x_3 both are complements of x_2 .

x_3, x_2 both are complements of x_1 .

⑧ Distributive property:-

$$a * (b + c) = (a * b) + (a * c)$$

$$a + (b * c) = (a + b) * (a + c)$$

⑨ Boolean Algebra:- A boolean algebra is a complemented distributive lattice. $\langle B, 0, 1, *, +, ' \rangle$

Let, $\langle P(A), \cap, \cup, \bar{}, \emptyset, A \rangle$

$$(P \vee Q) \wedge R \rightarrow (P \vee R)$$

$$((\neg P \wedge \neg Q) \vee R) \vee (P \vee R)$$

$$[(\neg P \vee \neg R) \wedge (\neg Q \vee \neg R)] \vee (P \vee R)$$

$$[\neg P \wedge (\neg Q \vee \neg R)] \vee [\neg R \wedge (\neg Q \vee \neg R)] \vee (P \vee R)$$

$$(\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R) \vee (\neg R \wedge \neg Q) \vee (\neg R \vee \neg R) \vee (P \vee R)$$

$$(\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R) \vee (\neg R \wedge \neg Q) \vee (\neg R) \vee (P \vee R)$$

$\wedge, \sim$

$$(b) ((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\equiv [\sim((P \vee Q) \wedge R)] \vee (P \vee R)$$

$$\equiv [\sim(P \vee Q) \vee \sim R] \vee (P \vee R)$$

$$\equiv (\sim P \wedge \sim Q) \vee (\sim R) \vee (P \vee R)$$

$$\equiv (\sim P \wedge \sim Q) \vee (\sim R \vee R) \vee P$$

$$\equiv (\sim P \wedge \sim Q) \vee (T \vee P)$$

$$\equiv (\sim P \wedge \sim Q) \vee T$$

$$\equiv (\sim P \wedge \sim Q)$$

$$((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\equiv (\sim((P \vee Q) \wedge R) \vee (P \vee R))$$

$$((\sim P \wedge \sim Q) \vee \sim R) \vee (P \vee R)$$

$$[(\sim P \vee \sim R) \wedge (\sim Q \vee \sim R)] \vee (P \vee R)$$

$$[(\sim P \vee \sim R) \vee (P \vee R)] \wedge [\sim Q \vee \sim R]$$

$$[(\sim P \vee P) \vee (\sim R \vee R)] [\sim Q \vee \sim R]$$

$$P \vee T \wedge P [T \vee T] \wedge [\sim Q \vee \sim R] \vee (P \vee R)$$

$$[T] \wedge [(\sim Q \vee \sim R) \vee (P \vee R)]$$

T

$$((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\sim[(P \vee Q) \wedge R] \vee (P \vee R)$$

$$(\sim(P \vee Q) \vee \sim R) \vee (P \vee R)$$

$$((\sim P \wedge \sim Q) \vee \sim R) \vee (P \vee R)$$

$$(\sim P \vee \sim R) \wedge (\sim Q \vee \sim R) \vee (P \vee R)$$

$$(\sim P \vee \sim R) \wedge (\sim Q \vee \sim P) \vee (R \vee \sim R)$$

$$[(\sim P \vee \sim R) \wedge (\sim Q \vee \sim P)] \vee T$$

$$\sim P \vee \sim R$$

$$((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\sim[(P \vee Q) \wedge R] \vee (P \vee R)$$

$$((\sim P \wedge \sim Q) \vee \sim R) \vee (P \vee R)$$

$$[(\sim P \vee \sim R) \wedge (\sim Q \vee \sim R)] \vee (P \vee R)$$

$$[\sim P \wedge (\sim Q \vee \sim R) \vee \sim R \wedge (\sim Q \vee \sim R)]$$

$$(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R)$$

$$((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\equiv \neg((P \vee Q) \wedge R) \vee (P \vee R)$$

$$\equiv \neg(P \vee Q) \vee \neg R \vee (P \vee R)$$

$$\equiv (\neg P \wedge \neg Q) \vee (\neg R \vee P) \vee R$$

$$\equiv (\neg P \wedge \neg Q) \vee (P \vee \neg R) \vee R$$

$$\equiv (\neg P \wedge \neg Q) \vee P \vee (\neg R \vee R)$$

$$\equiv [(\neg P \wedge \neg Q) \vee P] \vee T$$

$$\equiv \cancel{[(\neg Q \wedge \neg P) \vee P] \vee T}$$

$$\equiv \cancel{\neg Q}$$

$$\equiv [P \vee (\neg P \wedge \neg Q)] \vee T$$

$$\equiv [(P \vee \neg P) \wedge (P \vee \neg Q)] \vee T$$

$$\equiv [T \wedge (P \vee \neg Q)] \vee T$$

$$\equiv (P \vee \neg Q) \vee T$$

$$\equiv P \vee \neg Q$$

$$(\wedge, \neg), (\wedge, \neg)_{P \vee Q} \quad \begin{array}{l} P \rightarrow Q \\ \neg P \vee Q \\ \neg Q \rightarrow \neg P \end{array}$$

$$((P \vee Q) \wedge R) \rightarrow (P \vee R)$$

$$\equiv \neg((P \vee Q) \wedge R) \vee (P \vee R)$$

$$[[(\neg P \wedge \neg Q) \vee R] \vee (P \vee R)]$$

$$[(\neg P \vee R) \wedge (\neg Q \vee R)] \vee (P \vee R)$$

$$[(\neg Q \vee R) \wedge (\neg P \vee R)] \vee (P \vee R)$$

$$(\neg Q \vee R) \wedge (\neg P \vee P) \vee (\neg R \vee R)$$

$$(\neg Q \vee R) \wedge (T) \vee (T)$$

$$(\neg Q \vee R) \wedge T$$

$$(\neg Q \vee R) \wedge (T \wedge R)$$

$$(\neg(P \vee Q) \wedge R) \vee (P \vee R)$$

$$(\neg P \wedge \neg Q) \vee (\neg R) \vee (P \vee R)$$

$$\cancel{(\neg P \wedge \neg Q) \vee (\neg R \vee P) \vee (\neg R \vee R)}$$

$$(\neg P \vee \neg R) \wedge (\neg Q \vee \neg R) \vee (P \vee R)$$

$$\cancel{(\neg P \vee \neg R) \wedge (\neg Q \vee \neg R)}$$

$$\begin{array}{l} \textcircled{1} \quad p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

$$\begin{array}{l} \textcircled{2} \quad p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

$$\begin{array}{l} \textcircled{3} \quad p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

$$\begin{array}{l} \textcircled{4} \quad p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

$$\begin{array}{l} \textcircled{5} \quad p \vee q \\ \neg q \\ \hline \therefore p \end{array}$$

$$\begin{array}{l} \textcircled{5} \quad p \\ p \vee q \\ \hline q \end{array}$$

$$\begin{array}{l} \textcircled{6} \quad p \\ p \wedge q \\ \hline q \end{array}$$

$$\begin{array}{l} \textcircled{7} \quad p \wedge q \\ \hline \therefore p \end{array}$$

$$\begin{array}{l} p \wedge q \\ \hline \therefore q \end{array}$$

$$\begin{array}{l} \textcircled{8} \quad p \vee q \\ p \rightarrow r \\ q \rightarrow r \\ \hline \therefore r \end{array}$$

Any CPU work different instructions with different CPU

For a given program executed on specific machine, we can define the following parameters:

(A) Instruction count - Total no. of instructions to be executed

(B) CPI - Cycles per instruction

(C) Clock cycle (H_z) = (also related to computational power-clock)

Execution Time: $IC * CPI * CLK$

Discrete Mathematics

Graph Theory $G(V, E)$

17/01/2023

• Directed Graph

• Undirected Graph $v_1 \rightarrow v_2$

If direction is not then

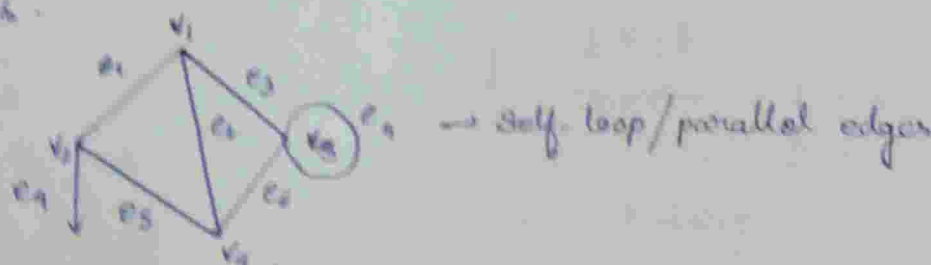
$$= \begin{cases} v_1 = N v_1 \\ v_2 = N v_2 \end{cases}$$

• Euler's Graph (Start $\Rightarrow$ End)

• Hamilton

• TSP Problem

$\rightarrow$ Edges are incident on vertices and vertices are incident on edges.



Pendant vertex (Hinge)
(dependent on only one vertex)

$\rightarrow$ Graph with no loop - Multi graph

$\rightarrow$ Graph with at least one loop - Pseudo graph

Discrete Mathematics:-

Graph Theory $G(V, E)$

19/01/2023 .

• Directed Graph

• Undirected Graph 

If direction is not then

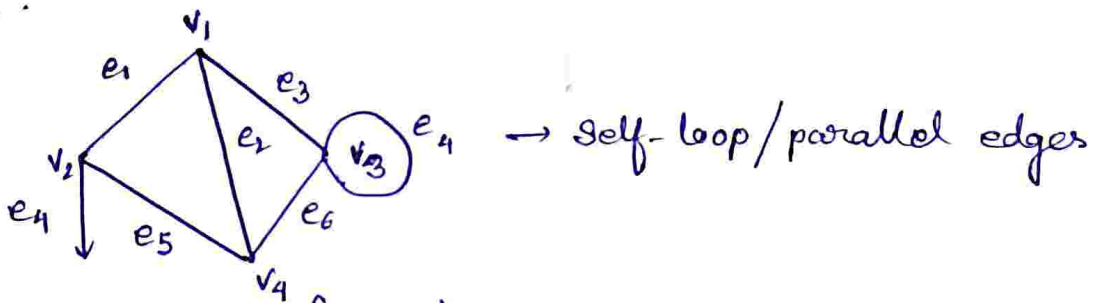
• Euler's Graph (Start $\Rightarrow$ End)

$$= \begin{cases} v_1 = N v_1 \\ v_2 = N_2 \end{cases}$$

• Hamilton

• TSP Problem

→ Edges are incident on vertices and vertices are incident on edges .



Pendant vertex (Hinge)
(dependent on only one vertex)

→ Graph with no loop - Multi graph

→ Graph with at least one loop - Pseudo graph .

- Graph with no loop and no parallel edges → Simple graphs
- Graph is not geometry, it is a pictorial representation.

• UAIPLY

$[N_1]$ $[N_2]$ $[N_3]$

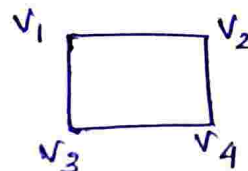
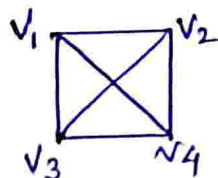
(A) (E) (W)

- Planar: Graph which can be drawn on plane.
- Degree: Has many edges passes through vertices.
- Regular graph - All the vertices are of the same degree.



• k-regular graph -

Complete graph →



Graph without edges are possible but not without vertex.

$G(N, E)$ E can be empty.

All vertices of degree 1 are pendent.

All vertices of degree 0 are isolated.

≡ Total degree = $2 \times$ edges of graph.

$$\sum_{i=1}^n d(v_i) = 2e$$

DISCRETE MATH:- Graph Theory:-

21/01/23

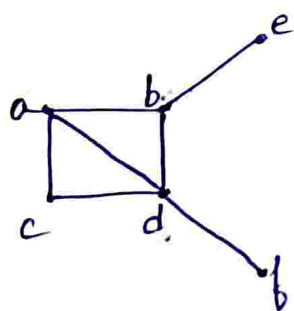
• The no. of vertices of odd degrees in a graph is even.

$$\sum_{i=1}^n d(v_i) = 2e = \sum_{\text{odd}} d(v_j) + \sum_{\text{even}} d(v_k)$$

• Walk - Alternating sequences of the vertices and edges in which we might repeat the vertices. (f-d-c-a-d-b)

• Path - Vertices will not be repeated in the path. (f-d-a-b)

• Closed path - start and end at the same vertices.
(a-b-d-c-a)



• Connected graph - We can reach any vertex from any vertex.

• Disconnected graph - If we can't reach any vertex from any vertex.

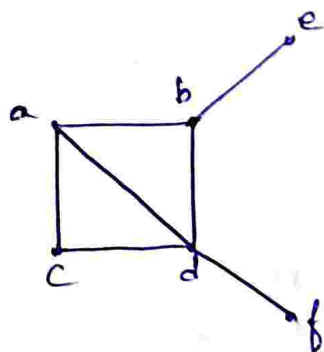
• Length of path - no. of edges in a path.

• Components - A graph is disconnected if it has many components.

•

i = isolated point.

Example of a disconnected graph.

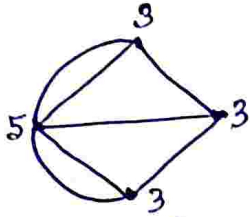


• Let, $G = (V, E)$ is a graph and R is a relation on V which is defined by $x R y$ iff $x = y$ or there is a path from x to y . Then we say, R is an equivalence relation for undirected graph.

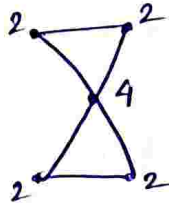
1 component = connected graph.

many components = equivalence classes = disconnected graph.

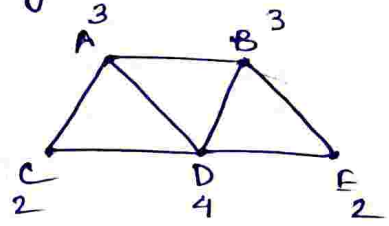
- Euler Graph:- All the vertices of even degree.



Not a Euler graph



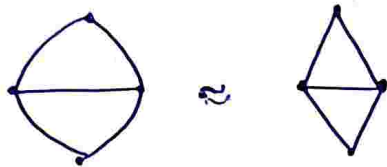
Euler graph.



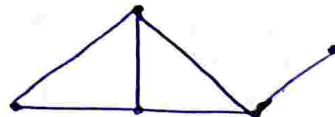
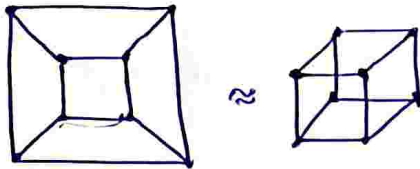
Unicursal graph.

[Open Euler line is, Start point $\neq$ End point]

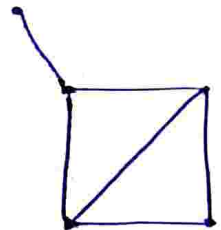
Book Narshingh Deo Graph Theory:



Isomorphic graph.
One-to-one mapping.

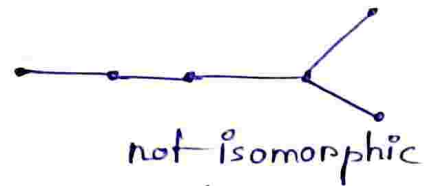


$\approx$

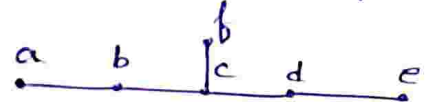


Conditions for isomorphic graph:-

- No. of vertices same
- No. of edges same
- No. of degrees of vertices same.
- Exactly one-to-one correspondence between the vertices.

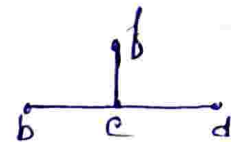


not isomorphic



1st Level of Isomorphism:-

- Sub-graphs:- Part of a graph
Sub-set of vertices and the edges connecting those vertices.

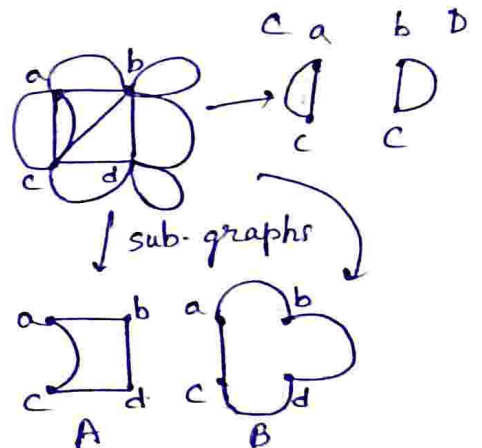


(i) Edge disjoint sub-graph:-

A and B have no edge in common so, they are edge disjoint subgraph.

(ii) Vertex disjoint sub-graph:-

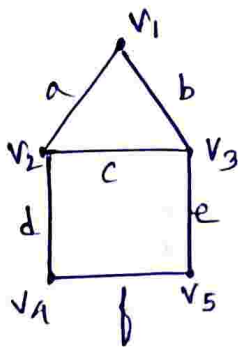
C and D have no vertex in common, so, they are edge vertex disjoint subgraph.



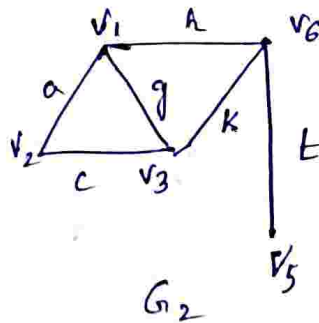
- Vertex disjoint subgraph is always edge disjoint but not the other way around.
- We define graph using two sets V and E , so set operations are valid on graph such as Union and Intersection.

e.g. ① $G_1 = (V_1, E_1)$ $G_3 = (V_3, E_3) = G_1 \cup G_2$
 $G_2 = (V_2, E_2)$ where $V_3 = V_1 \cup V_2$
 $E_3 = E_1 \cup E_2$

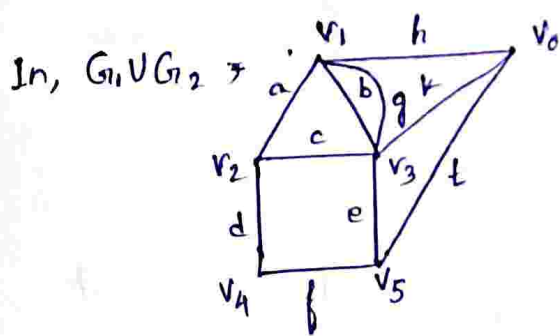
② $G_4 = (V_4, E_4) = G_1 \cap G_2$ where, $V_4 = V_1 \cap V_2$ (common vertices)
 $E_4 = E_1 \cap E_2$ (common edges)



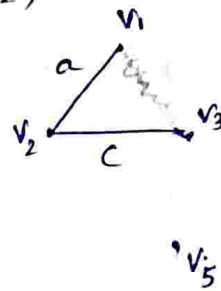
G_1



G_2



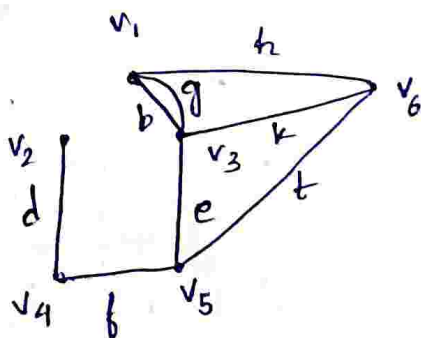
In, $G_1 \cap G_2$



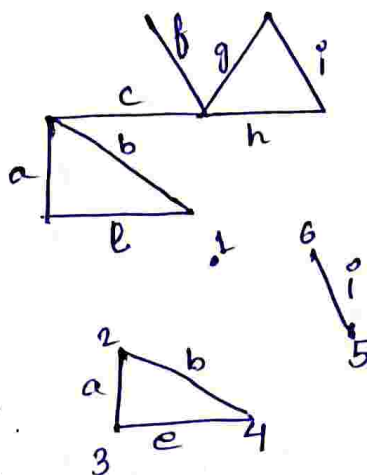
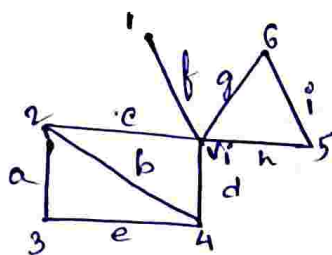
• Ring sum operation: $G_5 = G_1 \oplus G_2 = (V_5, E_5)$

$V_5 = V_1 \cup V_2$

$E_5 = E$ exclude the common edges.



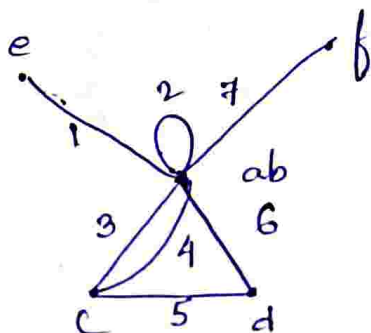
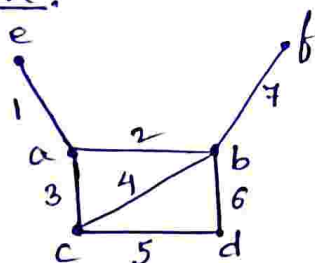
- Det Deletion:- (Deletion of edges or vertices).



Deletion of edge is like $G-d$.

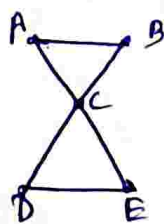
Deletion of vertex is like $G-v_i$.
(deleting vertex along with its incident edges).

- Fusion:

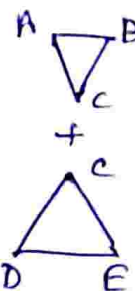


Merger of two vertices.

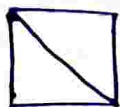
- Theorem:- A connected graph is an Euler graph iff it can be decomposed into circuits.



Edge disjoint sub-graphs

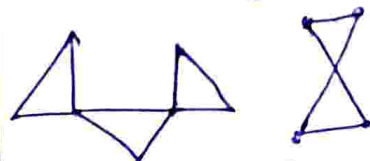
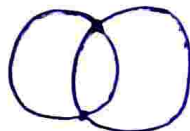
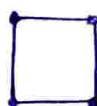


} Form circuits.

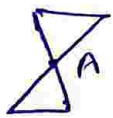


Open Euler line,
not an Euler graph
hence doesn't form
edge disjoint circuit.

- Arbitrarily traceable graph:-



Not arbitrarily
traceable as many
possibilities to trace
path that might not
cover all vertex.



This graph is arbitrarily traceable ~~for~~ from node A.

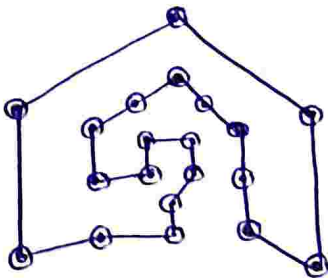
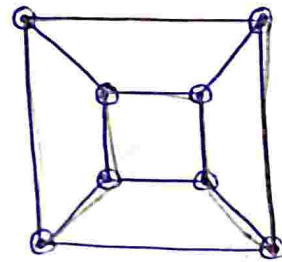
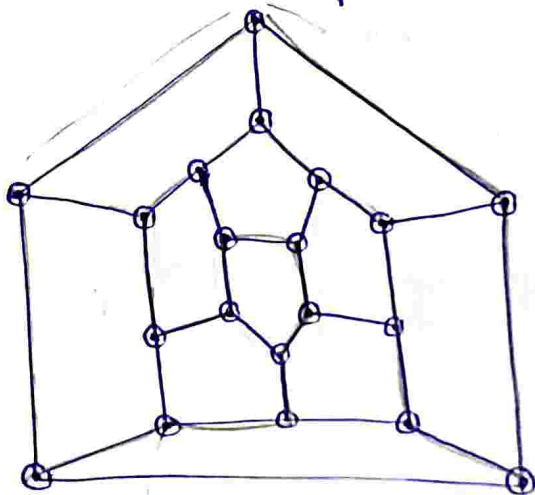
A graph is arbitrarily traceable if the vertex is present in every circuit.

* Discrete Mathematics:-

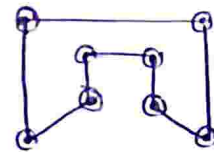
- 02/02/2023
- ① What is the difference between macros and function?
 - ② Is it main recursive? ③ Complete graph
 - ③ How to compare string?

* Hamiltonian Graphs - Vertices are not to be revisited.

• 20 cities connected by hamiltonian circuit.



Hamiltonian circuit

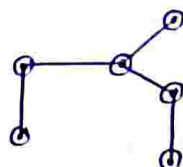


Hamiltonian circuit.

- In a complete graph of 3 or more vertices, hamiltonian circuit must form.
- Complete graph - a graph in which all vertices are connected to each other.

- Tree - Tree is a ^{connected circuitless} cycle-less graph.

e.g.

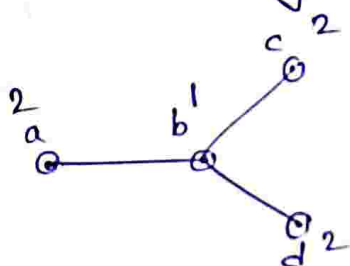


- Forest - A collection of trees i.e., collection of several components that are all trees.

- A graph that is connected graph with n vertices and $(n-1)$ edges is called forms a tree.

- If in a graph if there is ~~only~~ one and only one path between any two vertices, then it is a tree.

- Tree is a minimally connected graph.



b is the centre of this tree.

- Eccentricity:- To determine the centre of a graph we calculate the eccentricity of the graph vertices.

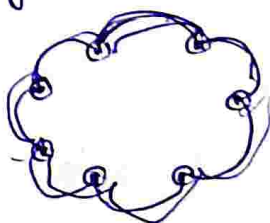
$E(v)$ = The distⁿ max-distance of v

$$= \text{Max}(v, v_i),$$

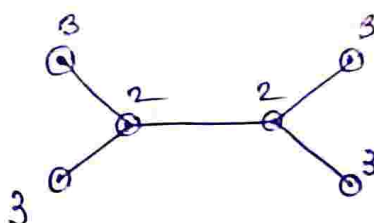
where v_i is the farthest vertex from v .

- Centre: The centre of graph is vertex of minimum eccentricity.

- In ring topology all vertices are centres.



- In a tree there are a maximum of two centres.



- Another method to find centre in a tree is to remove the pendent vertices and deduce graph such that only ~~or~~ 1 or 2 vertices are left.

• Radius: The eccentricity of the ^{centre} vertex $\uparrow$ is the radius of the centre graph.

• Diameter: Max. eccentricity is the diameter of the graph.

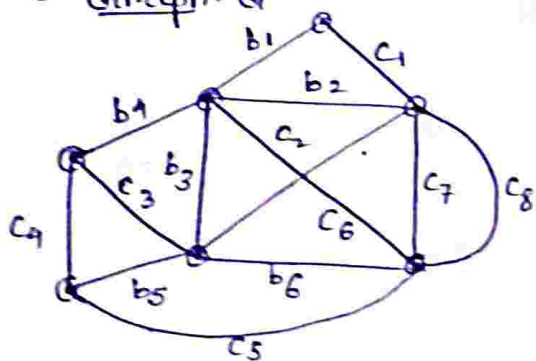
• All trees are not rooted trees. If a tree has a vertex that has some special property then that vertex is called the root of the tree and the graph is called rooted tree.

• Tree derived from a graph: In a graph with e edges, total sub-graphs possible is 2^e . In these sub-graphs some are trees. These trees are derived from a graph.

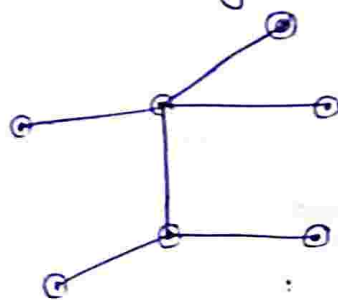
• In such ~~trees~~ sub-graphs, there is a tree that has all vertices of the graph, then that tree is called a spanning tree. (skeleton) (maximal tree)

A spanning tree is not independent, it is always derived from some graph. There can be more than one spanning tree.

• Graph-G



Spanning tree:



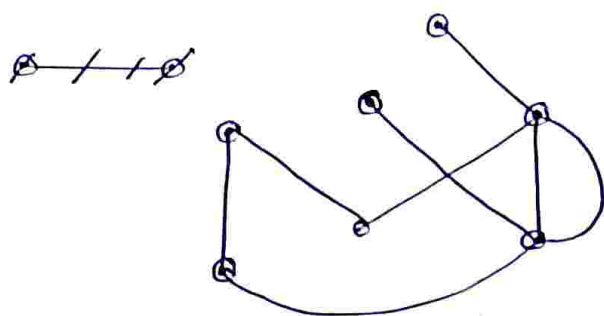
• Properties of spanning tree:-

(i) ~~Branch~~ Edges included in a spanning tree are called branches.

(ii) Edges not included in a spanning tree are called chords.

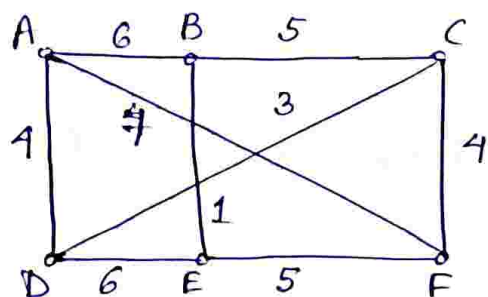
(iii) No. of branches + No. of chords = Total no. of edges in graphs.
 $B + C = E$

- Co-tree/Complementary tree:- The tree graph obtained w.r.t to the chords. It is also not unique, but no. of branches is intact, so are total no. of chords.



- In case of complete graphs, no. of spanning trees is $n^{(n-2)}$ trees.

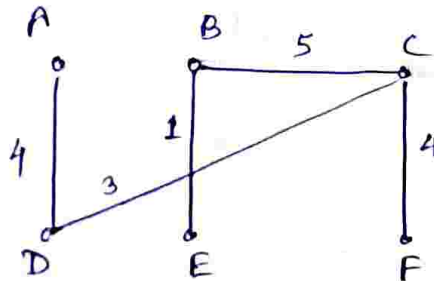
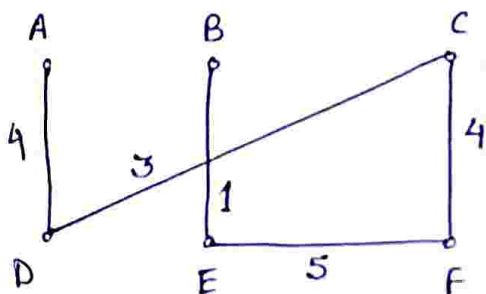
- Prims and cruscal Algorithm: To obtain minimum spanning tree.



- Cruscal Algorithm:-

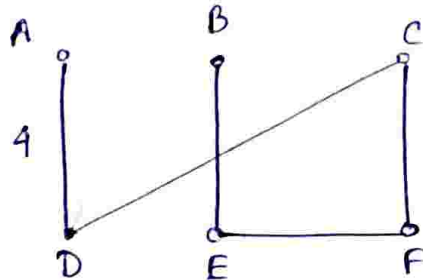
① Sort weights: 1, 3, 4, 4, 5, 5, 6, 6, 7
(Stable sort)

② Write the vertices and join edges with least weight one by one until spanning tree is obtained.



Minimum spanning trees

- Prims Algorithm:-

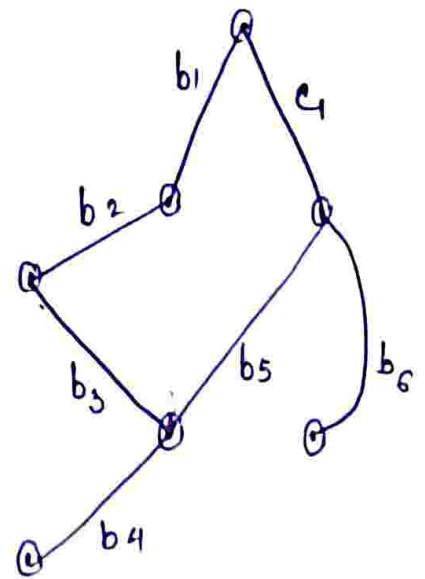
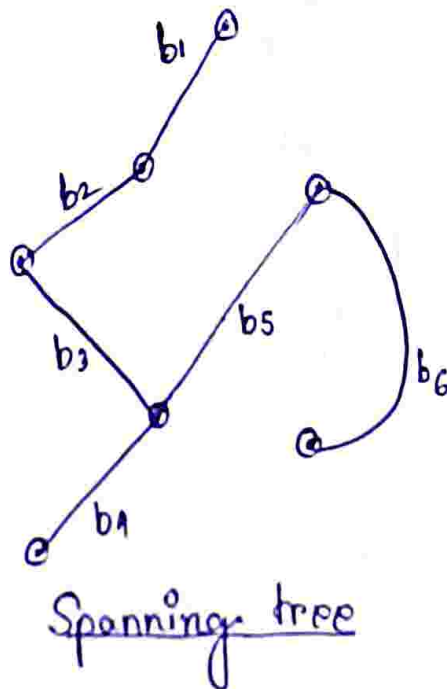
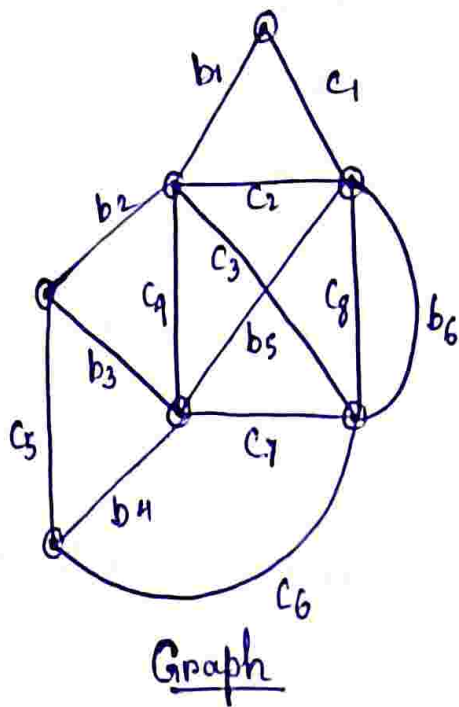


	A	B	C	D	E	F
A		6		④		7
B	6		5		1	
C		5		3		④
D	4		③		6	
E		①		6		5
F	7		4		⑤	

Discrete Mathematics:-

14/02/2023

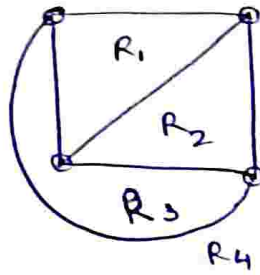
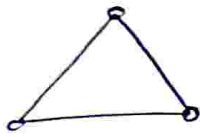
① Fundamental circuit : Circuit obtained by adding a chord to a spanning tree, is called fundamental circuit.



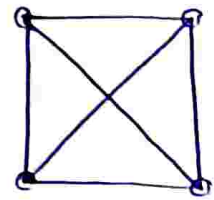
② There are as many fundamental circuit as there are chords

③ Planar graph:- Graph that can be drawn in a plane and edges ~~are~~ don't intersect to each other is called planar graph.

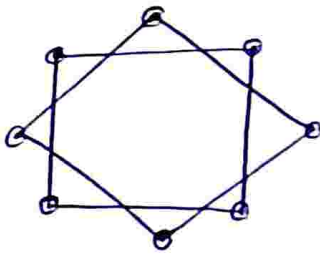
Examples:-



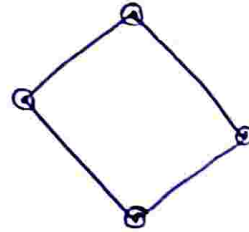
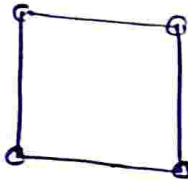
Isomorphic to



Hence, planar.



22



$$\therefore v=4$$

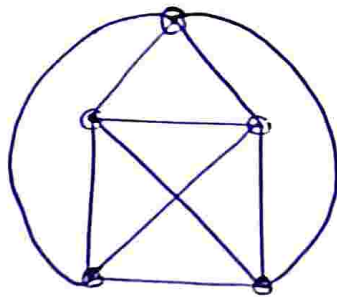
$$e=6$$

$$r=4$$

$$\therefore v-e+r=4-6+4=2$$

Two components of same graph, hence planar.

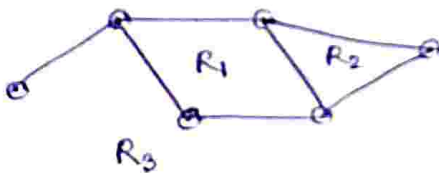
Non-planar:-



① Thickness of graph - No. of planes required to draw a graph.

② If any graph satisfies the the property $v-e+r=2$, then it is said to be planar graph. This formula was given by Euler.
 v = vertices, e = edges and r = regions/faces.

Examples-

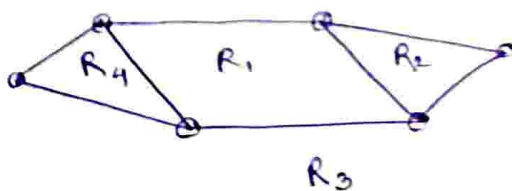


$$v=6$$

$$\text{Hence, } 6-7+3=2$$

$$e=7$$

$$r=3$$

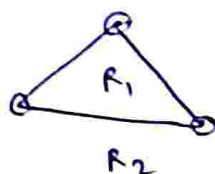


$$v=6$$

$$\text{Hence, } 6-8+4=2$$

$$e=8$$

$$r=4$$



$$v=3$$

$$\text{Hence, } 3-3+2=2$$

$$e=3$$

$$r=2$$

$$v - e + r = v - 1 - (e - 1) + r = v - 1 - e + 1 + r = v - e + r = 2.$$

Proof of the formula by induction,

$$v - e + r = v - (e - 1) + (r - 1) = v - e + 1 + r - 1 = v - e + r = 2.$$

For, $n=1$, then ~~for~~ assume for $n=k$ and prove it for $n=k+1$

⊙

$$v=1$$

$$e=0$$

$$r=1$$

For, ① Pendant edge $e=e-1, v=v-1, r=r$

② Non pendant $e=e-1, v=v, r=r-1$

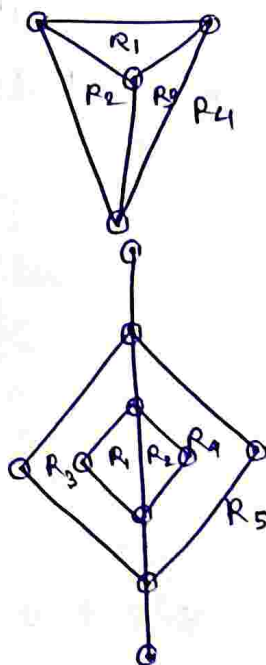
$$\therefore v - e + r = 1 + 1 = 2.$$

• No. of vertices = v

No. of edges = e

No. of edges in spanning tree = $v-1$

$\therefore$ No. of region, $r = e - (v-1) + 1$



$$v = 4$$

$$e = 6$$

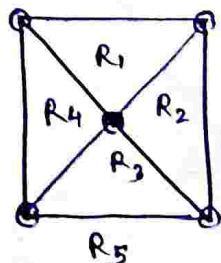
$$r = 4$$

$$v - e + r = 4 - 6 + 4 = 2$$

Hence, planar.

$$v - e + r = 10 - 13 + 5 = 15 - 13 = 2$$

Hence, planar.



$$v - e + r = 5 - 8 + 5 = 10 - 8 = 2$$

Hence, planar.

• $v - e + r = 2$ is the necessary condition but not the sufficient one.

Kuratowski's Graph:-

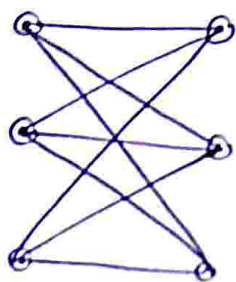
Th 1 K_n is planar iff $n \leq 5$ (i.e., a complete graph can be obtained till $n \leq 5$).

- Bipartite graph (two partition of vertices):- নিজস্ব সাথে connected নয়
কিন্তু অন্য vertex-এর সাথে
connected.



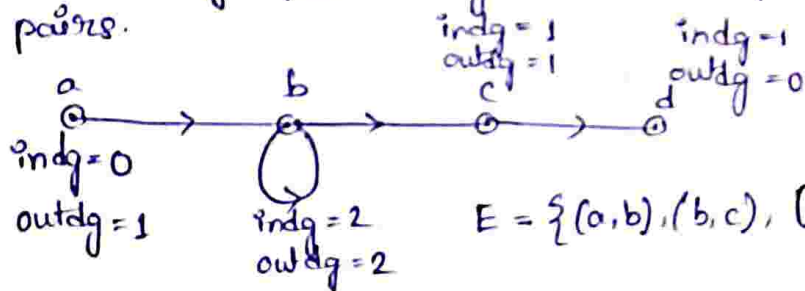
- Complete bipartite graph by Kuratowski:-

Th 2 $K_{m,n}$ (where m and n are the no. of vertices in the two sets) is planar iff $m \leq 3$ or $n \leq 3$.



$K_{3,3}$

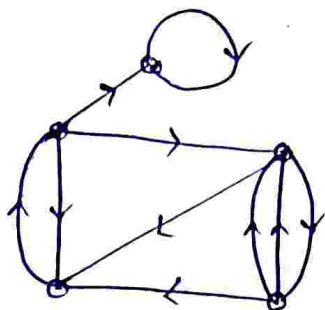
- Directed graphs:- Edges can be represented in terms of ordered pairs.



$$E = \{(a,b), (b,c), (c,d), (b,b)\}$$

- If we attach direction, then graph is called oriented the graph.
- If we remove direction from a directed graph, then it the graph is called underlying graph.
- On (a,b) vertices a and b are incident.
- On vertex c edge (b,c) is incident.
- Indegree - outgoing incoming no. of edges. (d^-)
- Outdegree - outgoing no. of edges. (d^+)

e.g.



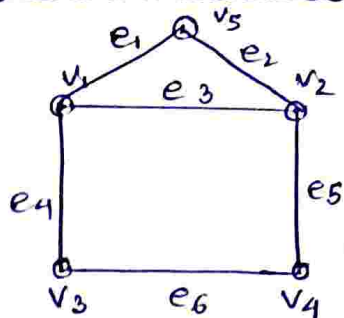
• edges are parallel when direction orientation is same.

• for pendant edge either indegree 1 or outdegree 1.

$$\text{i.e., } d^+(v) + d^-(v) = 1$$

■ Incidence Matrix and Adjacency Matrix:-

Incidence matrix: $V \times E$



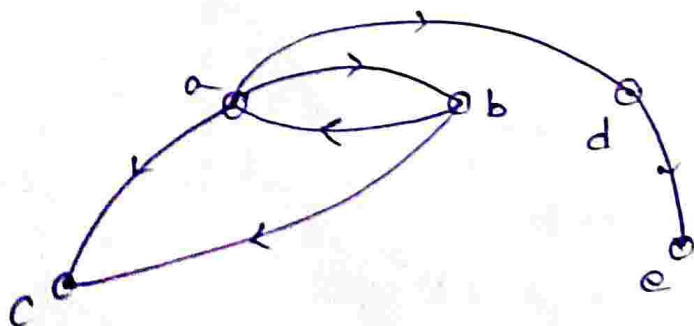
$v \backslash e$	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	0	1	1	0	0
v_2	0	1	1	0	1	0
v_3	0	0	0	1	0	1
v_4	0	0	0	0	1	1
v_5	1	1	0	0	0	0

Adjacency matrix: $V \times V$

$v \backslash v$	v_1	v_2	v_3	v_4	v_5
v_1	0	1	1	0	1
v_2	1	0	0	1	1
v_3	1	0	0	1	0
v_4	0	1	1	0	0
v_5	1	1	0	0	0

example:- $V = \{a, b, c, d, e\}$

$E = \{(a, b), (b, a), (a, c), (a, d), (b, c), (d, e)\}$



	a	b	c	d	e
a	0	1	1	1	0
b	1	0	1	0	0
c	0	0	0	0	0
d	0	0	0	0	1
e	0	0	0	0	0

- Weighted adjacency matrix.
- Shortest path program
- Dijkstra's algorithm

Adjacency matrix:

Indegrees:-

$$a = 1$$

$$b = 1$$

$$c = 2$$

$$d = 1$$

$$e = 1$$

(column-wise)

Outdegrees:-

$$a = 3$$

$$b = 2$$

$$c = 0$$

$$d = 1$$

$$e = 0$$

(Row-wise)

Computer Fundamentals:-

13/02/2023

